

NCRB CRIME DATA ANALYSIS

End-to-End Data Analytics & Business Intelligence Project

Technology Stack	Excel • Python • Pandas • SQL • Power BI • DAX
Analysis Period	2017–2022
Dataset Structure	Approximately 5,920 records and 31 columns
Analysis Level	Year • State • District • Crime Category
Dashboard	5-page interactive Power BI report

Project Overview

This report documents the complete workflow used to transform NCRB crime data into an analysis-ready dataset and an interactive Power BI dashboard. The project demonstrates practical skills in data preparation, SQL, data handling, DAX, visualization, and analytical reporting.

1. Abstract

This project presents an end-to-end analysis of National Crime Records Bureau (NCRB) crime data covering 2017 to 2022. The objective was to transform a structured crime dataset into useful analytical views for examining reported cases across time, states, districts, and crime categories.

The workflow begins with the source dataset in Excel, followed by data inspection and cleaning using Python and Pandas. The prepared data is analyzed using SQL. SQL transformations provide analysis-ready datasets for Power BI. Power BI is used to build a five-page dashboard, while DAX is used for dynamic KPIs and year-over-year calculations.

The final solution provides executive, category, state, district, and year-over-year analysis in a single portfolio project.

2. Introduction

Crime data contains multiple geographical levels and numerous crime classifications. Raw tabular data can be difficult to interpret directly because users must aggregate and compare records manually. Data analytics tools can transform these records into structured views that make patterns easier to explore.

This project was developed as a practical data analytics portfolio project combining spreadsheet data, Python programming, SQL analysis, and Power BI visualization.

3. Problem Statement

The source dataset contains multiple years, states, districts, registration information, and many crime-related fields. A raw-data user would have to manually calculate trends, comparisons, and category totals. The project therefore aims to create a reusable analytical workflow and dashboard.

Key questions

- How do reported cases change over the years?
- Which states and districts have the highest reported case volumes?
- Which crime categories have the largest reported case volumes?
- How do selected crime categories change over time?
- How did reported cases change from one year to the next?

4. Objectives

- Clean and validate the source data using Python and Pandas.
- Use SQL for data aggregation, transformation, and analysis.
- Create reusable analytical views for year, state, district, category, and YoY analysis.
- Create DAX measures for important dashboard KPIs.
- Build a five-page interactive Power BI dashboard.
- Present the results as a professional data analytics portfolio project.

5. Dataset Description

The working dataset is represented by a table named **crime_records**. It covers 2017–2022 and contains approximately 5,920 records and 31 columns.

Core identification fields

Field	Description
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year	Year of the recorded data
state_name / state_code	State identification
district_name / district_code	District identification
registration_circles	Administrative/registration grouping

Crime fields

The dataset contains 25 crime-related measures, including murder/homicide, rape/sexual violence, attempted rape, assault on modesty, kidnapping/abduction, human trafficking, crimes against children, POCSO crimes, cruelty/domestic violence, dowry crimes, acid attack, suicide abetment, hurt/grievous hurt, juvenile crimes, cybercrime, cyber fraud, identity/privacy crimes, cyber harassment/threats, online sexual/obscene crimes, cheating/fraud, theft/property crimes, forgery/counterfeiting, immoral traffic/prostitution, and other IPC/special laws.

Because some fields represent related or subdivided categories, the project does not blindly add every raw crime column into a single universal total. Aggregations are based on the prepared analytical views.

6. Methodology

The project follows a sequential data analytics workflow from source data to reporting.

Excel Source data	Python Cleaning	SQL Transformation	Power BI Visualization	DAX KPIs
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6.1 Data Preparation — Python

Python and Pandas were used to inspect the source data, review structure and data types, examine missing values, and perform data-quality checks. Duplicate-looking records were investigated at the value level rather than automatically removed. Records with the same location identifiers but different crime values were treated as distinct records.

6.2 SQL Analysis

SQL was used to aggregate records and create reusable analytical views for downstream reporting. A long-format view named **vw_crime_long** was created with year, state, district, registration circle, crime category, and crime count.

6.3 Power BI

Power BI was used as the reporting layer. The final report contains five pages designed to move from high-level KPIs to category, state, district, and time-based analysis.

7. Python Data Preparation

- Import the Excel dataset using Pandas.
- Inspect row count, column count, column names, and data types.
- Check missing values and potential duplicate records.
- Validate year, state, district, and crime fields.
- Prepare the cleaned dataset for SQL analysis.

Example Python workflow

```
import pandas as pd

df = pd.read_excel("crime_records.xlsx")
df.head()
df.info()
df.isnull().sum()
```

```
df.duplicated().sum()
```

8. SQL Analysis

SQL provides the main transformation and analytical layer. Queries were used to create reusable views rather than rebuilding the same calculations separately inside every Power BI visual.

Main analytical views

View	Purpose
vw_crime_long	Long-format crime-category analysis.
vw_pbi_yearly	Year and crime-category analysis.
vw_pbi_state	State-level analysis.
vw_pbi_district	District-level analysis.
vw_pbi_yoy	Year-over-year analysis.

Example SQL aggregation

```
SELECT year, state_name, SUM(total_cases) AS total_cases
FROM vw_pbi_yearly
GROUP BY year, state_name
ORDER BY year, total_cases DESC;
```

9. Power BI and DAX

Power BI is the interactive visualization layer. The report contains KPI cards, charts, slicers, tables, and DAX measures.

Important DAX measures

Total Reported Cases

```
SUM('public vw_pbi_yearly'[total_cases])
```

Latest Year

```
FORMAT(MAX('public vw_pbi_yearly'[year]), "0")
```

YoY Growth %

```
DIVIDE(CurrentYearCases - PreviousYearCases, PreviousYearCases, 0)
```

Average Cases per District

```
AVERAGEX(VALUES('public vw_pbi_district'[district_name]), CALCULATE(SUM('public vw_pbi_district'[total_cases])))
```

These measures make the KPI layer dynamic and demonstrate practical DAX usage rather than relying only on static numbers.

10. Dashboard Pages

10.1 Executive Overview

The Executive Overview provides the high-level summary. KPI cards display Total Reported Cases, Latest Year, Highest Crime State, Year-over-Year Growth, and Average Cases per District. Supporting visuals include yearly crime trend, state comparison, crime-category analysis, top districts, cybercrime trends, and a year selector.

10.2 Crime Category Analysis

This page focuses on crime-category distribution and yearly trends. The trend chart uses the **sum of total cases**, because a row count measures records rather than reported cases. A crime-category slicer allows the user to focus the trend analysis on selected categories.

10.3 State Analysis

The State Analysis page compares reported cases across states and provides a state-level crime-category breakdown. It supports comparison of overall volume and category composition.

10.4 District Analysis

The District Analysis page provides district-level comparison and a detailed district table. The prepared district view contains 24 districts, so the final page uses a direct comparison rather than an additional Top-N restriction.

10.5 Year-over-Year Analysis

The Year-over-Year Analysis page provides a year selector and a detailed table containing current cases, previous-year cases, and growth percentage. The final design prioritizes readability over an overcrowded multi-line chart.

11. Key Findings

- **Total Reported Cases:** approximately 8.66 million in the dashboard KPI.
- **Latest Year:** 2022.
- **Highest Crime State:** Tamil Nadu in the dashboard's state-level KPI.
- **Year-over-Year Growth:** approximately -6.4% from 2021 to 2022.
- **Average Cases per District:** approximately 10.99K.
- **Coverage:** analysis is available across years, states, districts, and 25 crime categories.

12. Technical Skills Demonstrated

Area	Skills
Data Handling	Excel, tabular data inspection, validation
Python	Python, Pandas, exploratory data analysis
SQL	Aggregation, GROUP BY, filtering, views, analytical transformations
Power BI	Dashboard design, slicers, charts, tables, formatting
DAX	Dynamic KPIs, YoY growth, averages, category ranking logic
Data Storytelling	Executive KPIs, trends, comparisons, analytical reporting

13. Limitations

- The dashboard describes reported case counts and should not be interpreted as a direct measure of the underlying incidence of crime.
- Crime categories can have related or hierarchical meanings; raw category fields should not automatically be summed without understanding their definitions.
- The analysis is limited to the fields and years available in the provided dataset.
- Results depend on the prepared dataset and aggregation logic used in the analytical views.

14. Future Scope

- Add population-normalized crime rates where reliable population data is available.
- Add geographic maps for state and district analysis.
- Add forecasting for selected crime categories.

- Automate the data pipeline from source data through SQL and Power BI.
- Introduce a formal star-schema model for larger-scale analytical workloads.
- Publish and schedule refreshes through Power BI Service where appropriate.

15. Conclusion

The NCRB Crime Data Analysis project demonstrates a complete analytics workflow from source data preparation to SQL transformation and interactive business intelligence reporting. By combining Excel, Python/Pandas, SQL, Power BI, and DAX, the project demonstrates practical skills beyond simple visualization.

The five-page dashboard provides a structured way to explore overall reported cases, crime categories, states, districts, and year-over-year changes. The project can therefore serve as a portfolio demonstration of data cleaning, SQL analysis, KPI design, dashboard development, and analytical storytelling.

16. References

1. National Crime Records Bureau (NCRB) — source dataset/documentation associated with the data used in this project.
2. Microsoft Power BI documentation — Power BI visualization and DAX concepts.
3. Python/Pandas documentation — tabular data processing and analysis.